Theorem [Birkhoff-von Neumann]:

Every $M \in \mathscr{B}_n$ can be written as a convex combination of permutation matrices:

$$M = \sum_{k=1}^{K} \lambda_k P_k, \quad \lambda_k \geq 0, \quad \sum_k \lambda_k = 1.$$

#3D251F cocap	#3DA2B7 teal	#ED6758 salmon	#FEC44C citrus	#D9B37E beige	#AA664A copper

Key Takeaway

Couplings generalize maps: Monge $\subset$ Kantorovich. This relaxation guarantees existence and convexity.

Setup

$$\mathscr{B}_n = \{M \in \mathbb{R}^{n\times n} \mid M\mathbf{1} = \mathbf{1}, M^\top \mathbf{1} = \mathbf{1}, M_{ij} \geq 0\}$$

(the Birkhoff polytope: set of $n \times n$ bistochastic matrices)

Birkhoff-von Neuman Theorem

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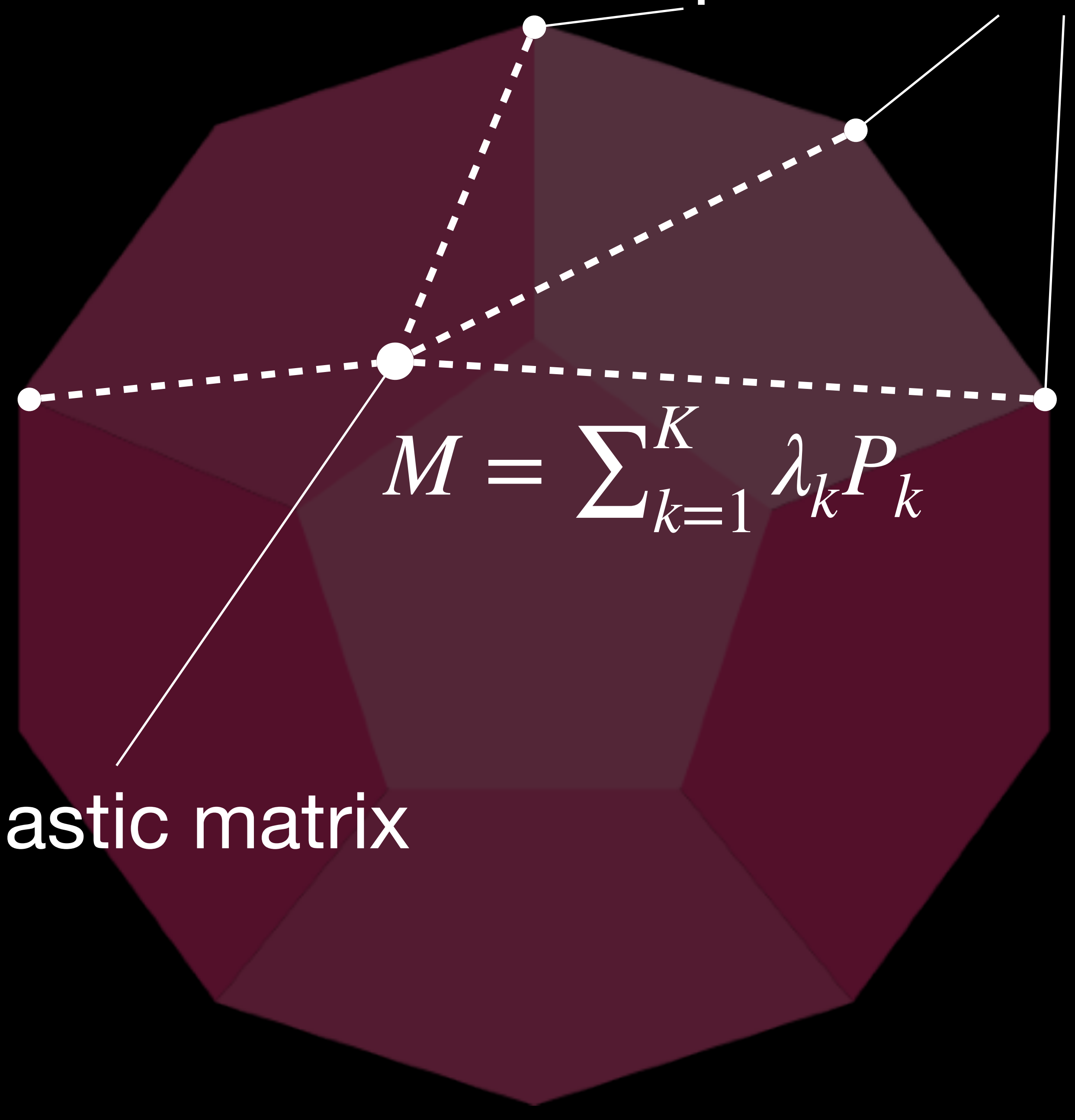


Implications for OT:

- Permutation matrices = Monge maps
- Bistochastic matrices = Kantorovich couplings
- Thus, Kantorovich is a convex relaxation of Monge.

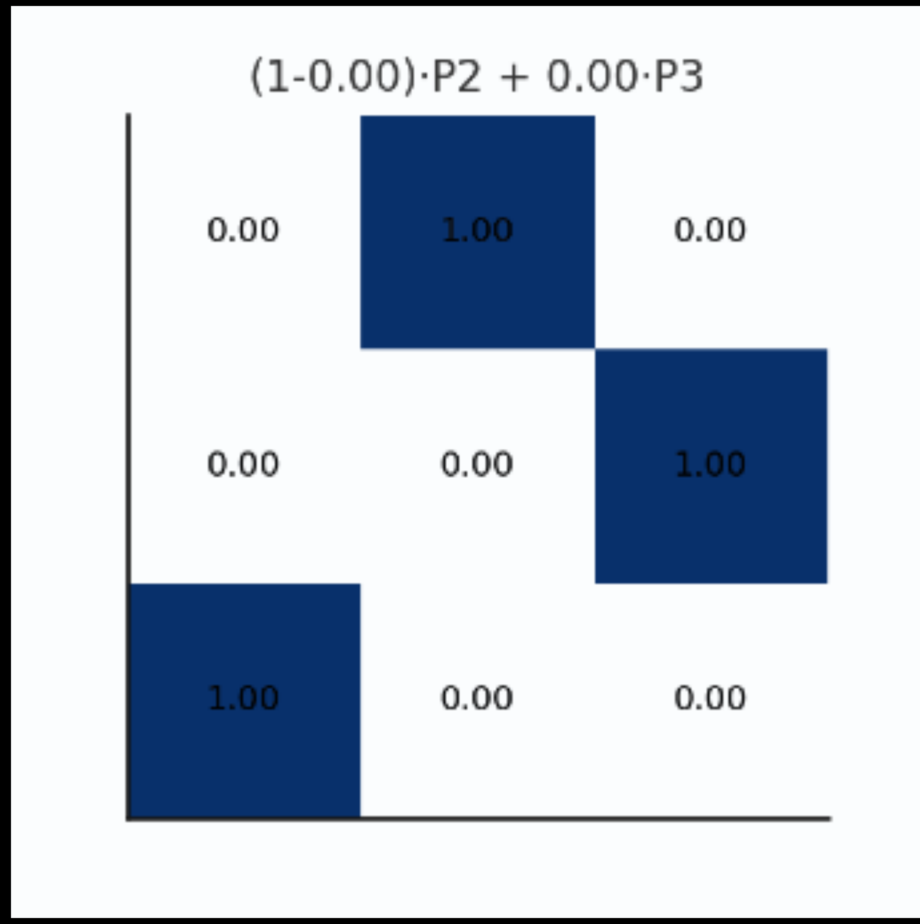
Key Takeaway

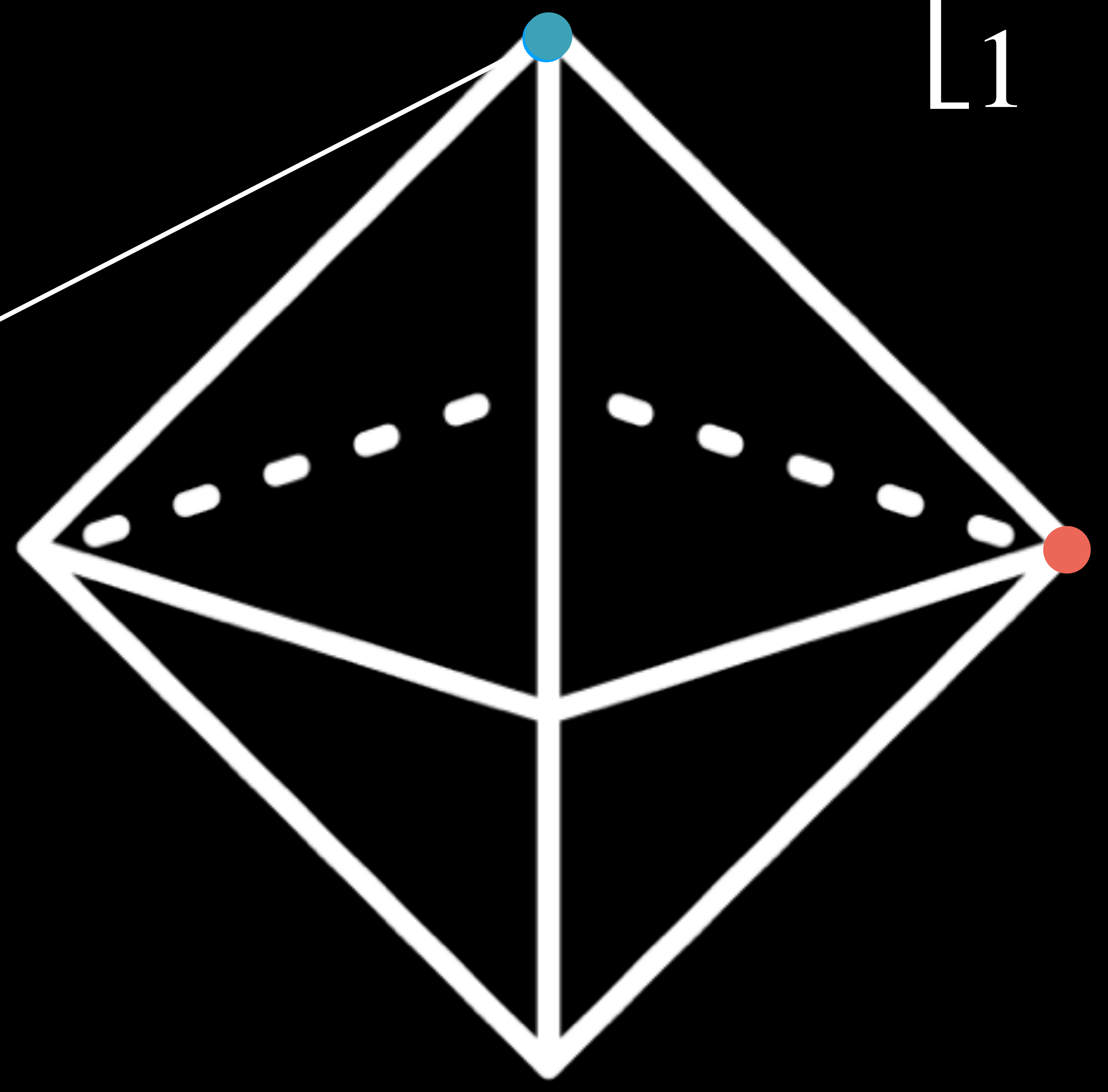
Couplings generalize maps:

Monge $\subset$ Kantorovich. This relaxation guarantees existence and convexity.

Couplings as Permutation Interpolation

$$P_2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$





$$P_3 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$M(t) = (1-t)P_2 + tP_3 = \begin{bmatrix} 0 & 1-t & t \\ t & 0 & 1-t \\ 1-t & t & 0 \end{bmatrix}$$